

1. For a complex Klein Gordon field consider the commutator $[\phi(x), \phi^\dagger(y)]$, and show that it vanishes for space-like separations.
2. Write the commutator in the form:

$$[\phi(x), \phi^\dagger(y)] = D(y - x) - \bar{D}(x - y), \quad (1)$$

where, $D(x - y)$ is the amplitude for a particle to propagate from x to y and $\bar{D}$ is the propagation amplitude for the antiparticle. Show that $D(y - x)$ is nonvanishing outside the light cone $|\mathbf{x} - \mathbf{y}| > |\mathbf{x}^0 - \mathbf{y}^0|$ and similarly for $\bar{D}$ (Hint: for this part you may wish to look at a similar calculation in Peskin and Schroder, Section 2.4, subsection **Causality**). Thus causality is maintained even though the particles and antiparticles propagate outside the light cones. The noncausal behaviour of particles going "forward" is cancelled by that of antiparticles going "backwards".

3. Following the same procedure discussed in class for the real scalar field, construct the Feynman propagator for the complex scalar field,

$$i\Delta_F(x_1 - x_2) = \langle 0 | T \phi(x_1) \phi^\dagger(x_2) | 0 \rangle.$$